

Workshop Guide

PyRosetta Physics Filters

Post-Design Validation

FastRelax >> Rosetta Filters >> Score Selection >> 3D Visualization

Companion material for the Google Colab notebook

Workshop_PyRosetta_Filters_Colab

Version 1.0 - September 2026

Table of Contents

1. Introduction & Where This Notebook Fits
2. Prerequisites & Installation
3. Uploading & Extracting Design Results
4. Stage 1: FastRelax - Structural Optimization
5. Binding Energy Calculation
6. Stage 2: Rosetta Physics Filters (RosettaScripts)
7. Understanding Each Filter
8. Stage 3: Score Filtering & Candidate Selection
9. 3D Visualization of Best Designs
10. Downloading Results
11. Parameter Reference
12. Troubleshooting
13. Next Steps & References

1. Introduction & Where This Notebook Fits

This notebook is the second half of a two-notebook workshop. The first notebook (Workshop_RFD3_Colab_v4) generates binder designs using RFdiffusion3 + LigandMPNN/solMPNN + ColabFold and produces a ZIP file with predicted structures. This second notebook takes that ZIP as input and applies Rosetta-level physics validation to identify the most experimentally promising candidates.

The pipeline has three computational stages:

Stage	Tool	Input	Output
1 - Relax	FastRelax	ColabFold PDBs	Relaxed PDBs + binding energy
2 - Filters	RosettaScripts	Relaxed PDBs	ddG, SAP, CMS, SASA, BUH
3 - Selection	Pandas	combined_scores.csv	filtered_scores.csv

Why physics filters after ColabFold? ColabFold gives you confidence metrics (ipTM, pLDDT, RMSD) that measure how well the sequence folds into the intended structure. But they do not measure physical properties like binding affinity, solubility, or interface packing quality. Rosetta's energy function fills this gap by evaluating the structures with an all-atom physics-based force field.

Estimated runtime: 5-15 minutes per structure (FastRelax is the bottleneck). For 16 designs, expect ~1-3 hours on a Colab GPU.

2. Prerequisites & Installation

Cell 0.1 - Install Dependencies

Installs all required packages:

Package	Purpose
pyrosetta	Rosetta modeling engine (via pyrosetta-installer)
py3Dmol	Interactive 3D protein visualization
biopython	PDB file parsing
pandas / numpy / matplotlib	Data analysis and plotting

IMPORTANT: PyRosetta requires an academic license from RosettaCommons. The pyrosetta-installer will prompt for your username and password. Register at <https://www.rosettacommons.org/software/license-and-download>

The installer automatically downloads the correct PyRosetta wheel for your Python version (~1.5 GB). This takes 3-5 minutes on Colab.

3. Uploading & Extracting Design Results

Cell 0.2 - Upload & Extract

This cell presents a file upload widget. Upload the *_designs.zip file produced by the first workshop notebook. The code then:

1. Extracts all files from the ZIP archive
2. Locates PDB files inside the 03_predictions/ folder
3. Selects only relaxed structures (files with '_relaxed_' in the name)
4. If no relaxed structures exist (use_relax was False), falls back to using all prediction PDBs
5. Copies selected PDBs into a clean input_pdb/ working directory

Directory Structure After Extraction

```
/content/pyrosetta_filters/
input_pdb/          <- Selected PDBs for processing
results/
  01_relaxed/       <- FastRelax output PDBs
  02_filtered/      <- RosettaScripts output PDBs
binding_energies.csv <- Stage 1 results
combined_scores.csv  <- Stage 2 all Rosetta scores
filtered_scores.csv  <- Stage 3 filtered candidates
```

If you ran the first notebook with use_relax = True, the relaxed structures are preferred because they already went through Amber energy minimization, giving Rosetta a cleaner starting point.

4. Stage 1: FastRelax - Structural Optimization

Before computing physics-based metrics, each ColabFold prediction is relaxed with Rosetta's FastRelax protocol. This is essential because AlphaFold2 predictions, while accurate in overall fold, can have small steric clashes and suboptimal side-chain rotamers that would produce misleading energy values if scored directly.

What FastRelax Does

FastRelax alternates between side-chain packing (finding optimal rotamers) and all-atom energy minimization (adjusting coordinates to lower the energy). It repeats this cycle multiple times (default: 15 repeats) with gradually decreasing repulsive weights, which lets the structure settle into a deep energy minimum.

Key Settings

Setting	Value	Purpose
standard_repeats	15	Pack/minimize cycles (more = better convergence)
constrain_relax_to_start	True	Keeps backbone close to AF2 prediction
coord_constrain_sidechains	True	Prevents large side-chain rearrangements
set_bb(True)	-	Allow backbone movement during minimization
set_chi(True)	-	Allow side-chain torsion optimization

Chain Configuration (Cell 1.0)

You must specify which chain is the binder (peptide) and which is the target. In designs from the first workshop,

chain A is typically the binder and chain B is the target. The partners string (e.g. 'B_A') tells Rosetta how to separate chains for binding energy calculations.

5. Binding Energy Calculation

After relaxation, the notebook computes binding energy for each complex by the unbinding method:

1. Score the bound complex: $E(\text{bound})$
2. Separate the chains by translating the binder 100 Å away
3. Score the separated complex: $E(\text{unbound})$
4. Binding energy = $E(\text{bound}) - E(\text{unbound})$

A negative binding energy means the complex is more favorable than the separated chains -- the more negative, the stronger the predicted binding. Values are in Rosetta Energy Units (REU).

Interpreting Binding Energies

Binding Energy (REU)	Interpretation
< -30	Strong binding - promising candidate
-30 to -15	Moderate binding - worth investigating
-15 to 0	Weak binding - likely not functional
> 0	Unfavorable - complex is destabilized

These are approximate thresholds. Rosetta energies are not absolute free energies -- useful for ranking and comparing designs, not for predicting exact K_d values.

6. Stage 2: Rosetta Physics Filters (RosettaScripts)

Stage 2 applies a comprehensive RosettaScripts XML protocol to each relaxed structure. The protocol first minimizes the interface region in Cartesian space, then evaluates a panel of physics-based filters.

The XML Protocol (Simplified)

The protocol runs in this order:

1. Interface minimization (Cartesian): Optimizes residues within 14 Å of the interface.
2. ddG filter: Binding free energy with 5 repeats and extreme value removal.
3. SAP score: Spatial Aggregation Propensity of the binder chain (solubility proxy).
4. Contact molecular surface: Measures interface surface complementarity.
5. SASA filters: Buried surface area (total, hydrophobic, polar).
6. BUH: Buried Unsatisfied H-bonds at the interface.

7. Understanding Each Filter

ddG (Binding Free Energy)

The most important single metric. Measures the energetic difference between the bound complex and the separated chains. Uses the Cartesian score function (beta_nov16_cart) with constraints, 5 repeated measurements, and extreme value removal for robustness.

ddG (REU)	Quality
< -40	Excellent
-40 to -30	Good
-30 to -20	Fair
> -20	Poor

SAP Score (Spatial Aggregation Propensity)

Predicts how likely the binder is to aggregate (clump together) in solution. High SAP scores indicate exposed hydrophobic patches. Only the binder chain (chain A) is scored.

SAP Score	Solubility Prediction
< 20	Highly soluble
20 - 35	Acceptable
35 - 50	Marginal - may aggregate
> 50	Likely to aggregate

Contact Molecular Surface (CMS)

Quantifies shape complementarity between binder and target interface surfaces. Higher values mean better geometric fit. Accounts for 3D curvature match, not just area.

CMS	Interface Quality
> 400	Excellent complementarity
300 - 400	Good
200 - 300	Fair
< 200	Poor fit

Interface Buried SASA

Total solvent-accessible surface area buried upon complex formation. Larger values = bigger interface. Reported as total, hydrophobic, and polar components. A good interface has both hydrophobic (driving force) and polar (specificity) contacts.

BUH (Buried Unsatisfied H-bonds)

Counts hydrogen bond donors/acceptors buried in the interface but NOT making H-bonds. Each costs ~1-3 kcal/mol in real energy. Lower is better -- ideally < 5 for a small binder interface.

8. Stage 3: Score Filtering & Candidate Selection

After computing all physics metrics, this stage applies user-defined cutoffs to select the most promising candidates:

Parameter	Default	Direction	Rationale
ddg_cutoff	-30	Lower is better	Strong binding affinity
sap_cutoff	35	Lower is better	Good predicted solubility
cms_cutoff	300	Higher is better	Good interface complementarity

A design passes only if ALL three conditions are met simultaneously. Results are displayed as a color-coded table and saved to `filtered_scores.csv`.

When No Designs Pass

If no designs pass all three cutoffs, the notebook shows the top 5 by ddG anyway. Consider relaxing cutoffs: ddG < -15 and SAP < 50 are reasonable for a workshop setting.

9. 3D Visualization of Best Designs

Cell 4.0 renders interactive 3D views of structures that passed physics filters using py3Dmol:

- Rainbow cartoon: Binder chain (N-terminus blue to C-terminus red)
- Light blue cartoon: Target protein
- Metric labels: ddG, SAP, and CMS for each design

The viewer searches for PDBs in order: filter output (02_filtered/), relaxed (01_relaxed/), input (input_pdbs/). The `n_show` slider controls how many designs to display (default: 4, max: 12).

Controls

- Rotate: Left-click and drag
- Zoom: Scroll wheel
- Pan: Right-click and drag

10. Downloading Results

Cell 5.0 packages all outputs into a ZIP and triggers download:

```
pyrosetta_filter_results.zip
  binding_energies.csv      (Stage 1)
  combined_scores.csv       (Stage 2 - all metrics)
  filtered_scores.csv       (Stage 3 - selected candidates)
  01_relaxed/               (FastRelax output PDBs)
  02_filter_output/         (RosettaScripts output PDBs)
```


11. Parameter Reference

Chain Configuration (Cell 1.0)

Parameter	Default	Description
peptide_chain	A	Binder chain ID (the designed chain)
target_chain	B	Target protein chain ID
fastrelax_repeats	15	Number of relax cycles (1-30)

Physics Filter Cutoffs (Cell 3.0)

Parameter	Default	Range	Workshop Tier	Strict Tier
ddg_cutoff	-30	-100 to 0	< -15	< -40
sap_cutoff	35	0 to 100	< 50	< 25
cms_cutoff	300	0 to 1000	> 200	> 400

Visualization (Cell 4.0)

Parameter	Default	Description
n_show	4	Number of top designs to render (1-12)

12. Troubleshooting

PyRosetta installation fails

Requires a RosettaCommons academic license. Register at <https://www.rosettacommons.org/software/license-and-download>

No PDBs found after ZIP extraction

Ensure you uploaded the correct *_designs.zip. The ZIP must contain a 03_predictions/ folder with .pdb files inside it.

FastRelax takes too long

Reduce fastrelax_repeats to 5 for testing. For 16 designs at 15 repeats, expect 1-3 hours.

Chains not found / fold tree error

Chain letters must match your PDB. Open the PDB and check the chain column (column 22). ColabFold may use different letters.

ddG returns NaN or very large values

Chains could not be separated properly. Check that the partners string (e.g. B_A) matches the actual chain IDs in the PDB.

No designs pass the filters

Relax the cutoffs: try ddG < -15, SAP < 50, CMS > 150. Default cutoffs are stringent for workshop designs.

Out of memory during RosettaScripts

Cartesian minimization is memory-intensive. Restart runtime and process fewer structures at a time.

py3Dmol shows empty viewer

PDB may have unusual formatting. Check for ATOM records and that chain letters match your settings.

13. Next Steps & References

What to Do After This Notebook

1. Combine metrics: Merge ColabFold metrics (ipTM, pLDDT, RMSD) with Rosetta physics metrics (ddG, SAP, CMS) for comprehensive ranking.
2. Molecular dynamics: Run 50-100 ns MD simulations on top 3-5 candidates to assess stability over time.
3. Experimental validation: Order gene synthesis for top candidates. Express, purify, and test binding with SPR, BLI, or pull-downs.
4. Iterate: Promising designs become starting points for a focused second round with tighter contig constraints.
5. Optimize expression: Use SAP score to guide mutations that improve solubility without disrupting binding.

Key References

- [1] Alford et al. (2017). The Rosetta All-Atom Energy Function. J. Chem. Theory Comput. 13(6), 3031-3048.
- [2] Conway et al. (2014). Relaxation of backbone bond geometry. Protein Science 23(1), 47-55.
- [3] Norn et al. (2021). Protein sequence design by conformational landscape optimization. PNAS 118(11).
- [4] Chaudhury et al. (2010). PyRosetta. Bioinformatics 26(5), 689-691.
- [5] Fleishman et al. (2011). RosettaScripts. PLoS ONE 6(6), e20161.

Useful Links

- PyRosetta: <https://www.pyrosetta.org/>
- RosettaCommons: <https://www.rosettacommons.org/>
- SAP score: Lauer et al., J. Pharm. Sci. 2012